8 **(a)** Fig. 8.1 shows a circuit. The circuit is designed to switch on a night light when the surroundings are dark.

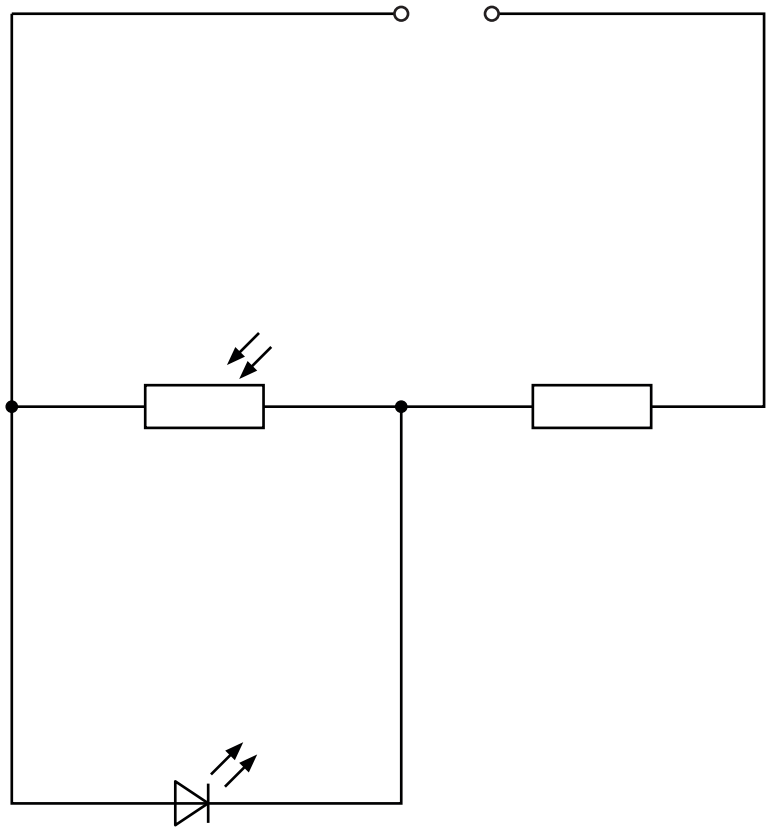

Fig. 8.1

(i) On Fig. 8.1, draw the circuit symbol for a voltmeter used to measure the potential difference (p.d.) across the light-dependent resistor (LDR). [1]

(ii) The surroundings change from light to dark.

1. State the effect of this change on the resistance of the LDR.

..... [1]

2. State and explain the effect of this change on the p.d. across the light-emitting diode (LED).

.....

.....

..... [2]

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(b) Fig. 8.2 shows another circuit. Lamps A and B are identical filament lamps.

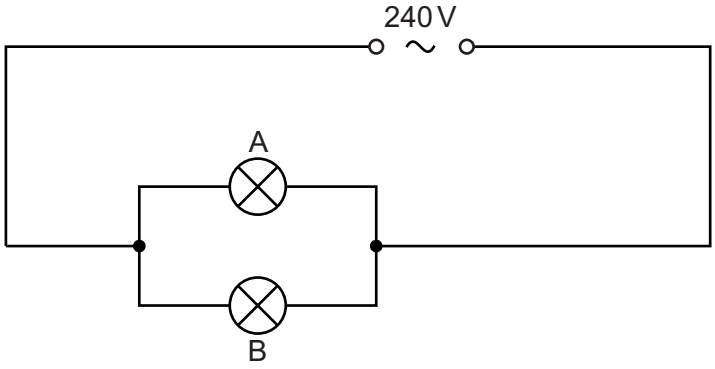

Fig. 8.2

The current supplied by the power supply is 0.50 A.

Calculate the resistance of lamp A.

resistance = ... [3]

[Total: 7]

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